

Vol 11 — Generative Geometry & Topology (BST Mathematical Foundations)

BST Physics Curriculum | Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate)

Status: SCAFFOLD v0.1 (Saturday 2026-05-23 Wave 2) — chapter content TO BUILD Lead: Lyra (primary, generative-geometry framing per Casey vision) + Grace (catalog backbone for Heegner + Monster + Stark deep-NT connections)

Reader target

Graduate mathematics + advanced physics. Vol 11 is the mathematical-foundations volume — explicitly NOT just “math methods for physics” but generative geometry + topology + deep number theory that BST has uncovered through substrate-cartography. Includes the Number-Theoretic Foundations chapter (Heegner + Monster + cyclotomic $Q(\zeta_{137})$ + Mersenne + Stark cross-link) which has become a major BST research thread.

Prerequisites

- Vol 0 Substrate Foundation (D_{IV}^5 + five integers)
- Vol 10 Math Methods (Lie groups + differential geometry + special functions)
- Graduate algebraic geometry + number theory background helpful (not required for Ch 1-3 motivation)

BST integer focus

Vol 11 is the substrate-cartography mathematical foundations volume. Documents the deep mathematical structures BST has uncovered: - Bounded Hermitian symmetric domains + Cartan classification + Helgason 1978 - Bergman + Faraut-Koranyi + Wallach representation theory + reproducing kernels - Heegner numbers (Stark 1967) + supersingular primes (Ogg 1975) + Monster moonshine (Borcherds 1992) - Cyclotomic $Q(\zeta_{137})$ + Mersenne primes + Reed-Solomon $GF(2^g)$ substrate-tick - K3 surfaces + Niemeier lattices + Leech lattice Bridge Object families - Generative geometry framing per Casey vision: D_{IV}^5 is generative geometric object that produces physics rather than describing it

Chapter outline (12 chapters, TO BUILD)

Ch	Title	Mathematical content	BST cross-link
1	Bounded Hermitian Symmetric Domains	Cartan classification (4 + 2 types); Helgason 1978 X.6.1	Vol 0 Ch 1 + Ch 9 Strong-Uniqueness
2	Bergman Reproducing Kernels	Bergman 1922 + Faraut-Koranyi 1994 normalization	T2442 $c_{FK} \cdot \pi^{(9/2)} = 225$ EXACT
3	Wallach K-Type Representation Theory	Wallach 1976 K-type classification of holomorphic discrete series	Vol 1 Ch 2 + Ch 5
4	Holomorphic Discrete Series + Highest Weights	Standard Lie-theory machinery	Casimir algebra Vol 1 Ch 5
5	D_IV ⁵ Geometry: Coset + Boundary + Coordinates	Type IV bounded domain realization (Cartan 1894 + Hua 1958)	Vol 0 Ch 1 + Ch 4 + Vol 1 Ch 2
6	K3 Surfaces + Niemeier Lattices + Leech	Hodge K3 + Mathieu + Leech lattice + Mathieu group	5-family Bridge Object architecture (Vol 0 Ch 9 C11)
7	Heegner Numbers + Cremona Elliptic Curves	Stark-Heegner 1967 class-number theorem; 49a1 + 27a1 + 121a1	Vol 0 Ch 9 C9 Stark anchor + Bridge Object family-members
8	Monster Moonshine + Supersingular Primes	Borcherds 1992 + McKay 1979 + Ogg 1975 + j-invariant	Monster supersingular $t_{\text{cosmo}} = 47 + 196883 = 47 \cdot 59 \cdot 71$ (Vol 4 Ch 4 T1924)

Ch	Title	Mathematical content	BST cross-link
9	Cyclotomic $Q(\zeta_N)$ + Reed-Solomon	Cyclotomic field theory + Galois groups + Reed-Solomon codes over $GF(2^g)$	K59 cyclotomic mechanism RATIFIED + Vol 1 Ch 10 substrate-tick
10	Mersenne Primes + Lucas-Lehmer	Mersenne prime theory + Lucas-Lehmer primality + density distribution	Sub-Substrate Mersenne Hierarchy T2451 + Vol 0 Ch 9 C15 (T2466 sub-result)
11	Generative Geometry Framing	D_{IV}^5 as generative object: substrate produces physics, not describes it	Casey Wednesday “ D_{IV}^5 as emergent TOE” + Vol 0 Ch 9 + Vol 4 Ch 9
12	Number-Theoretic Foundations Synthesis	Heegner + Monster + cyclotomic + Mersenne + Stark unified	Vol 0 Ch 9 Strong-Uniqueness 7-candidate enumeration

~50% existing BST coverage (substantial research exists; Heegner + Stark + Monster + Mersenne + cyclotomic are mature research lines); 50% to organize at curriculum-readable depth.

Cross-volume reference map

Inputs from Vol 0: - D_{IV}^5 + five integers + Strong-Uniqueness Theorem 7-candidate enumeration

Inputs from Vol 1: - Bergman + Wallach + Faraut-Koranyi anchors - Substrate-tick + cyclotomic mechanism

Outputs to: - All other volumes — Vol 11 is the deep mathematical foundation for BST’s number-theoretic claims - Number theorists + algebraic geometers joining BST research community

— Vol 11 INDEX v0.1, Saturday 2026-05-23 Wave 2 scaffold